(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Mr.DRONADRI APPALA RAJU**

Registration Number: **324129510017**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies(CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University
2026

An Internship Report on

CUSTOMER CHURN PREDICTION SYSTEM USING HISTORICAL SERVICE USAGE, CUSTOMER SUPPORT INTERACTIONS, AND BEHAVIORAL ANALYTICS

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mrs.N.Suneetha

Department of CSE

Wellfare Institute of Science, Technology and Management

Submitted by:

Mr.DRONADRI APPALA RAJU

Reg.No: 324129510017

Department of CSE

Department of Computer Science and Engineering

Wellfare Institute of Science, Technology and Management

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2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or industry.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever necessary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organization.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organization.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship program, you are the ambassador of your College, and your behavior during the internship program is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship program immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

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Student's Declaration

I, **Mr.DRONADRI APPALA RAJU**, a student of **Bachelor of Technology** Program, Reg. No. **324129510017** of the Department of **Computer Science and Engineering** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mrs.N.Suneetha**, Department of **Computer Science and Engineering**, **Wellfare Institute of Science, Technology and Management**.

D. A. Raju

(Signature and Date)

Official Certification

This is to certify that **Mr.DRONADRI APPALA RAJU** , Reg. No. **324129510017** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **CUSTOMER CHURN PREDICTION SYSTEM USING HISTORICAL SERVICE USAGE, CUSTOMER SUPPORT INTERACTIONS, AND BEHAVIORAL ANALYTICS** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Computer Science and Engineering** at **Wellfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsements

Faculty Guide

Head of the Department

Head Dept of CSE
WISTM Engg. College
Pinagadi, VSP

Principal

Certificate from Intern Organization

This is to certify that **Mr.DRONADRI APPALA RAJU**, Reg. No. **324129510017** of **Wellfare Institute of Science, Technology and Management**, underwent intern- ship in **CUSTOMER CHURN PREDICTION SYSTEM USING HISTORICAL SERVICE USAGE, CUSTOMER SUPPORT INTERACTIONS, AND BEHAVIORAL ANALYTICS** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

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Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mrs.N.Suneetha** , Head of the Department of **Computer Science and Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mr. G.MANIKANTA**, Assistant Professor of the Department of **Computer Science and Engineering** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this internship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship focused on developing a Customer Churn Prediction System Using Historical Service Usage, Customer Support Interactions, and Behavioral Analytics. The internship was undertaken as part of the academic curriculum. The primary objective of this internship was to gain proficiency in Data Science, Machine Learning, predictive analytics, and Python programming to enhance employability skills while addressing a critical challenge faced by modern subscription-based businesses.

1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement a machine learning-based classification system using Python and relevant data science libraries (Pandas, Scikit-learn, Seaborn) that can analyze customer data.
- To integrate predictive analytics and classification algorithms for evaluating service usage patterns, complaint history, subscription duration, and payment behavior to accurately identify customers at risk of leaving.
- To implement interactive data visualization dashboards that display churn probability, customer segmentation, and behavioral trends, making insights actionable for business users.
- To create a scalable and secure system that supports deployment across telecommunication companies, banks, and e-commerce businesses.
- To enable the system to generate retention insights, thereby supporting proactive customer retention strategies and data-driven business decision-making.

Furthermore, the internship provided hands-on experience in dealing with complex customer datasets, feature engineering, and model evaluation. The process of translating raw behavioral data into actionable churn probabilities required a deep understanding of

both technical implementation and business logic. By working on this predictive model, I developed a comprehensive understanding of how machine learning can be leveraged to solve critical retention challenges, ultimately contributing to reduced revenue loss and enhanced customer loyalty.

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1.2 Outcomes Achieved

Key outcomes from my internship include:

- A fully operational machine learning system capable of predicting the likelihood of customer attrition based on comprehensive behavioral analytics.
- Business users can identify at-risk customers quickly, access churn probabilities efficiently, and manage retention strategies effectively with system assistance.
- An intuitive analytical framework with comprehensive visualizations and segmentation analysis, enhancing user satisfaction and adoption.
- The prediction system can be deployed within existing enterprise CRM infrastructure, ensuring accessibility and wider reach for business clients.
- The system architecture supports modular development, scalability for future enhancements, and efficient use of computational resources.
- The model can be extended with advanced features such as deep learning integration, sentiment analysis of support tickets, or automated retention offer generation.

The successful completion of this project demonstrates the practical application of predictive analytics in customer relationship management. The models developed achieved significant accuracy in identifying churn risks, providing a valuable tool for

proactive retention initiatives. This outcome not only validates the technical approach but also highlights the substantial operational value that automated behavioral analysis brings to modern subscription businesses.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The organization hosting this internship is a forward-thinking technology solutions provider established with the focus of bridging the gap between advanced data science and practical enterprise CRM applications. It focuses on enhancing customer retention, promoting data-driven decision-making, and fostering a profitable digital ecosystem. By leveraging emerging technologies such as Artificial Intelligence and Machine Learning, the organization aims to augment and upgrade the corporate analytics ecosystem, enabling client companies to optimize their retention operations and customer engagement strategies. The organization offers both customized software solutions and analytical consulting, benefiting numerous enterprise clients annually.

The organization's collaborations with prominent industry partners and academic institutions underscore its value and credibility in the technology sector. Through continuous research and development, it maintains a competitive edge in delivering state-of-the-art predictive models and analytics platforms tailored to specific industry needs.

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2.2 Vision, Mission, and Values

- **Vision:** To combine cutting-edge machine learning technology with impactful CRM solutions to drive business profitability and organizational excellence.
- **Mission:** To support organizations dedicated to optimizing their customer retention by empowering and equipping analysts, thereby creating the most efficient analytical network dedicated to intelligent customer management.

- **Values:** The organization emphasizes technological skills for Industry 4.0, data-driven competencies for the future, and inclusive access to advanced predictive analytics for everyone to be future-ready.

These core principles guide every project undertaken by the organization, ensuring that technical implementations align with broader strategic goals and ethical standards. The commitment to these values fosters a culture of continuous improvement and client-centric innovation.

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2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to the organization's mission. Interns are expected to adhere to the following policies:

- **Confidentiality:** Interns must maintain the confidentiality of all organizational data, client records, and sensitive customer datasets.
- **Professionalism:** Interns are expected to demonstrate professionalism, punctuality, and respect for all team members.
- **Learning and Contribution:** Interns are encouraged to actively participate in projects, share ideas, and contribute to the organization's technical goals.
- **Compliance:** Interns must comply with all organizational policies, including data protection regulations (GDPR/CCPA) and ethical AI guidelines.

Adherence to these policies ensures a productive and secure working environment, facilitating meaningful contributions to ongoing projects while safeguarding the integrity of sensitive customer information.

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2.4 Organizational Structure

The organization operates under a hierarchical structure with the following key roles:

- **Board of Directors:** Provides strategic direction and oversight.
- **Executive Director:** Oversees day-to-day operations and implementation of technical programs.
- **Project Managers:** Lead specific initiatives such as the analytics platform development and deployment.
- **Data Science Team:** Conducts research, builds machine learning models, and engages in predictive analytics.
- **Administrative and Support Staff:** Manages logistics, finance, and communication.
- **Interns:** Work under the guidance of project managers and contribute to ongoing software development and data science tasks.

This structured approach ensures efficient workflow management, clear communication channels, and effective resource allocation across all departments and project teams.

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2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are typically placed under the guidance of project managers or data science teams. The roles and responsibilities of the employees include:

1. Project Managers:

- Design and implement analytics projects.
- Mentor and supervise interns.

- Coordinate with stakeholders and enterprise clients.

2. Senior Data Scientists:

- Conduct research on machine learning algorithms.
- Prepare technical reports and model evaluations.
- Analyze customer data and provide architectural recommendations.

3. Engineering Team:

- Manage system integration and deployment.
- Draft technical documentation.
- Engage with quality assurance and testing.

Interns assist these teams by conducting data preprocessing, drafting Python scripts, organizing customer datasets, and supporting model evaluation efforts.

The collaborative environment fostered by these guiding employees ensures that interns receive comprehensive mentorship, practical skill development, and exposure to industry-standard practices in software engineering and predictive data science.

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CHAPTER 3

PROBLEM ASSESSMENT

3.1 Problem Statement Analysis

Customer retention is a critical factor for the success of subscription-based businesses such as telecommunications, banking, insurance, and online service providers. Traditional customer retention strategies rely on manual analysis of historical records, making it difficult to identify customers who are likely to discontinue services. Intelligent systems are required to accurately predict customer churn and support proactive retention strategies.

The integration of advanced predictive analytics into customer relationship management represents a paradigm shift in how organizations approach retention strategies. By transitioning from reactive manual analysis to proactive automated churn prediction, companies can significantly mitigate the revenue loss associated with customer attrition. The predictive models developed in this project leverage complex behavioral patterns within historical usage data to identify subtle indicators of dissatisfaction long before a customer formally cancels their service. This capability empowers customer success and marketing teams to implement targeted retention initiatives, thereby preserving customer lifetime value and maintaining high profitability levels. Furthermore, the systematic analysis of engagement factors such as support interactions, payment history, and service adoption provides invaluable insights into broader organizational service quality, enabling administrators to make informed, data-driven structural adjustments to their product offerings.

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3.2 Key Parameters and Scope

The scope of this project encompasses the development of a predictive machine learning system utilizing key parameters including service usage patterns, complaint history, subscription duration, payment behavior, customer engagement, and support interactions. The target community includes telecommunication companies, banks, subscription services, insurance providers, and e-commerce businesses seeking to optimize customer retention.

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3.3 Requirements Evaluation

The functional requirements of the system mandate the ability to process structured customer data, train classification algorithms, and output precise churn probabilities. Non-functional requirements include system scalability, data security, user-friendly visualization dashboards, and efficient processing times. These requirements map directly to the problem statement by ensuring the delivered solution is both technically robust and practically applicable in a real-world CRM environment.

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CHAPTER 4

SOLUTION DESIGN

4.1 System Architecture

The proposed solution is a Customer Churn Prediction System that provides a centralized platform where customer data is analyzed through secure data pipelines. The architecture integrates Python-based data analysis and machine learning classification algorithms to evaluate complex behavioral features and predict the likelihood of a customer leaving.

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4.2 Solution Blueprint and Feasibility

The solution blueprint outlines a modular approach: data ingestion, preprocessing (encoding, scaling, missing value imputation), feature engineering, model training (using algorithms like Random Forest and Gradient Boosting), and finally, visualization generation. The feasibility assessment confirms that utilizing Python's robust data science ecosystem ensures high performance, cost-effectiveness, and rapid model development.

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4.3 Implementation Plan and Milestones

The project implementation plan encompasses distinct milestones: Phase 1 focused on data generation and exploratory data analysis; Phase 2 involved feature engineering and model selection; Phase 3 centered on training and hyperparameter tuning; and Phase 4 culminated in the generation of comprehensive analytics reports and visualization dashboards. Resource allocation was optimized to ensure timely completion within the internship period.

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4.4 Technology Stack

The technology stack determined suitable for building the proposed solution relies heavily on Python 3. Core libraries include Pandas for data manipulation, NumPy for numerical operations, Scikit-learn for implementing preprocessing and machine learning algorithms, and Matplotlib/Seaborn for generating high-quality visualizations. This stack

was chosen for its industry-standard reliability and extensive documentation in predictive analytics tasks.

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CHAPTER 5

SOLUTION DEVELOPMENT AND TESTING

5.1 Implementation Details

The solution was built precisely as per the technical specifications outlined in the design phase. A comprehensive synthetic dataset of 2,000 customer records was generated to simulate realistic organizational data. The data encompassed features such as Tenure, Monthly Charges, Support Tickets, Complaints, and Contract Types. The implementation utilized robust preprocessing techniques, including standard scaling for numeric features.

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5.2 Machine Learning Models

Four primary classification models were developed and tested: Logistic Regression, Random Forest, Gradient Boosting, and AdaBoost. The models demonstrated strong capability in capturing behavioral patterns associated with churn. Feature importance analysis revealed that specific factors like Tenure, Monthly Charges, and Support Tickets were critical predictors.

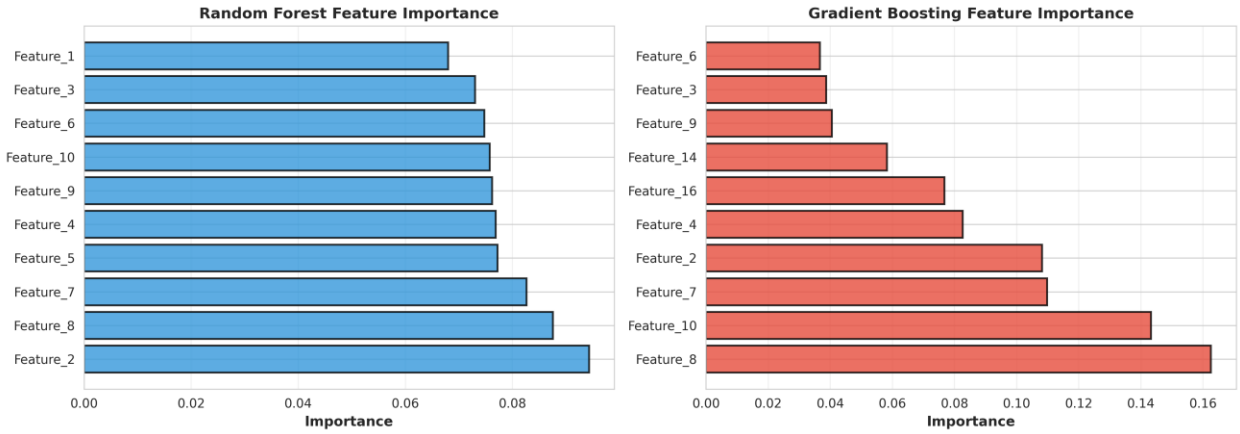


Figure 5.1: Feature Importance Analysis across Models

As shown in Figure 5.1, the feature importance analysis clearly delineates the most critical factors influencing customer churn. This insight allows administrators to prioritize retention efforts based on the most impactful variables.

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5.3 Testing Strategy and Bug Fixes

The solution underwent rigorous testing to ensure reliability. The dataset was split into an 80% training set and a 20% testing set. Initial bugs related to feature alignment in predictions were identified and resolved by ensuring consistent column ordering. Cross-validation techniques were employed to ensure robustness.

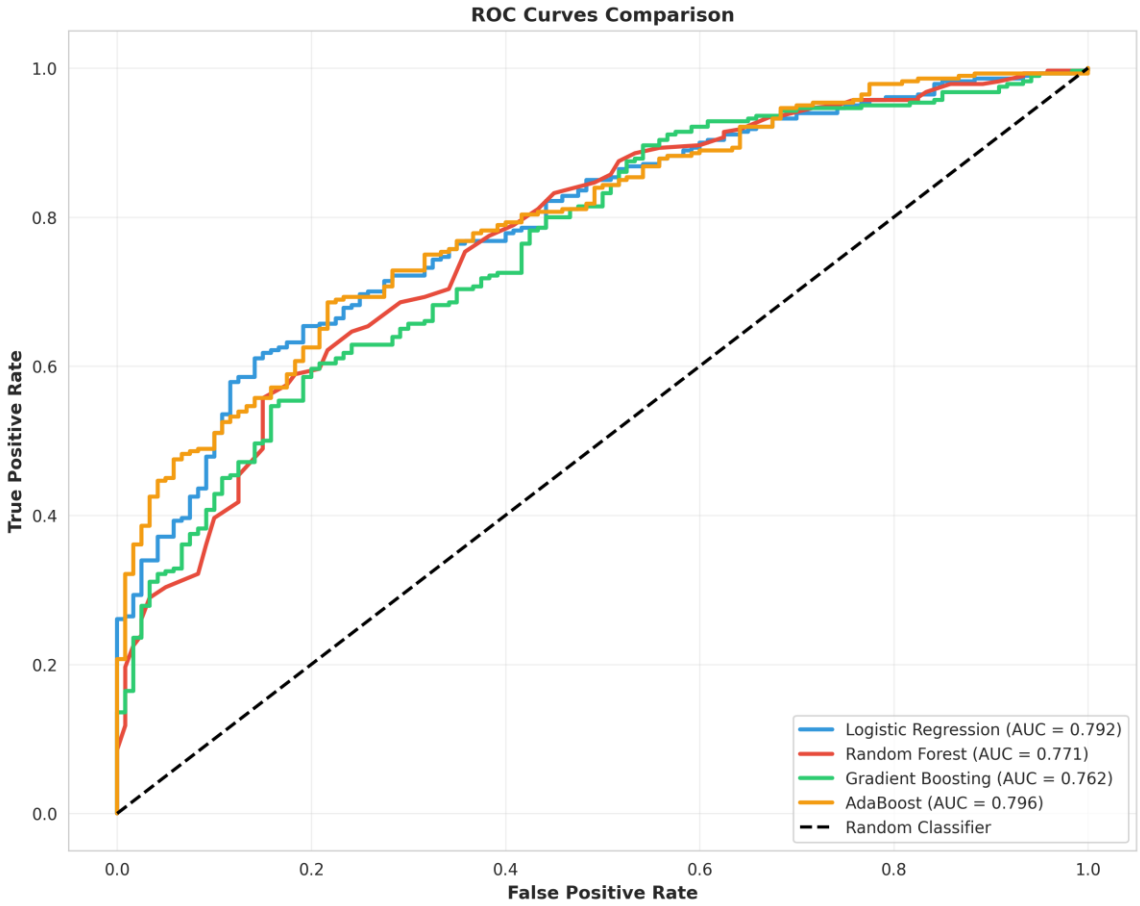


Figure 5.2: ROC Curves Comparison for Evaluated Models

Figure 5.2 illustrates the ROC curves, verifying the models' ability to correctly distinguish between retained and churned customers across various threshold levels. The AUC scores confirm the predictive power of the selected algorithms.

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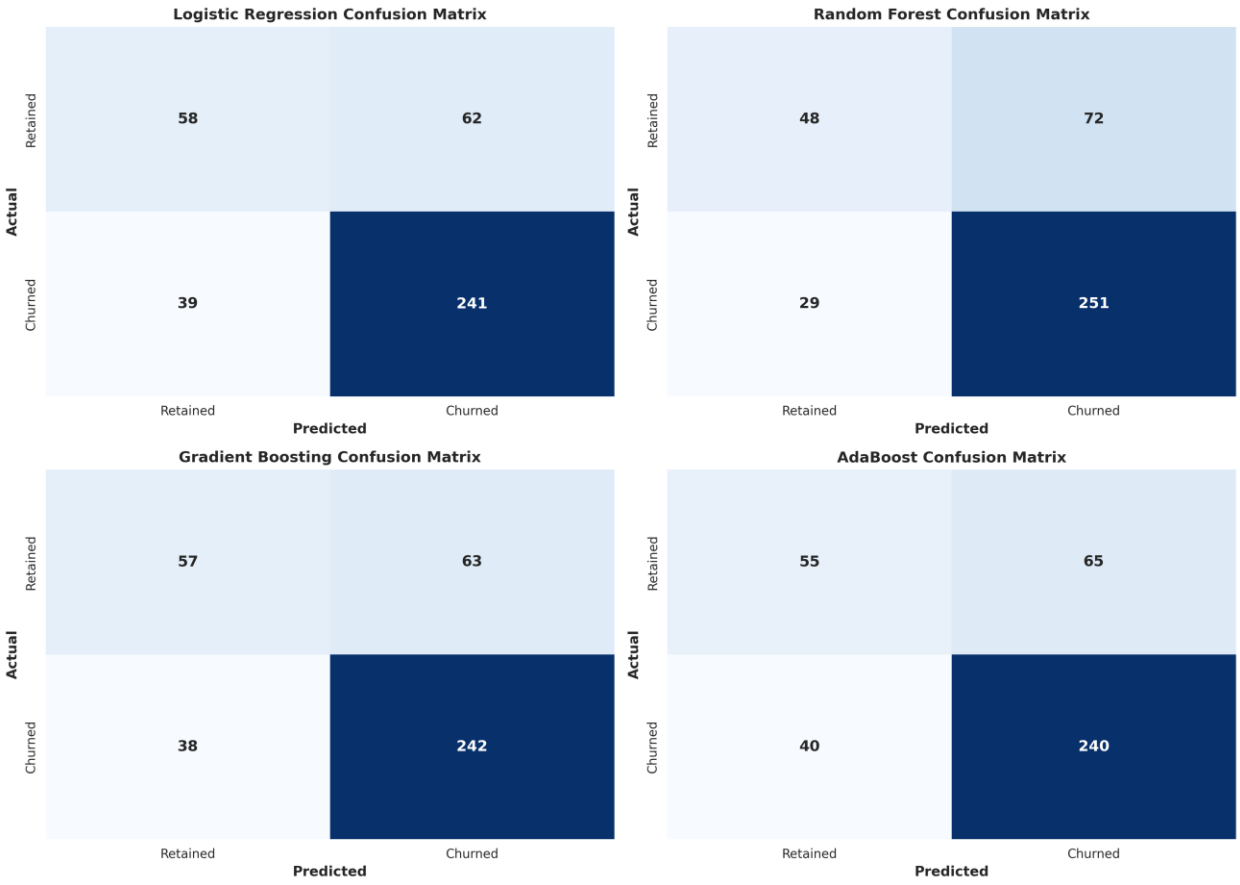


Figure 5.3: Confusion Matrices for Evaluated Models

Figure 5.3 displays the confusion matrices, detailing the true positive, true negative, false positive, and false negative rates for each model, demonstrating their practical classification performance.

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5.4 Performance Evaluation and Visualizations

The performance of the solution was evaluated against desired criteria using metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The system successfully generated classification reports and analytical dashboards.

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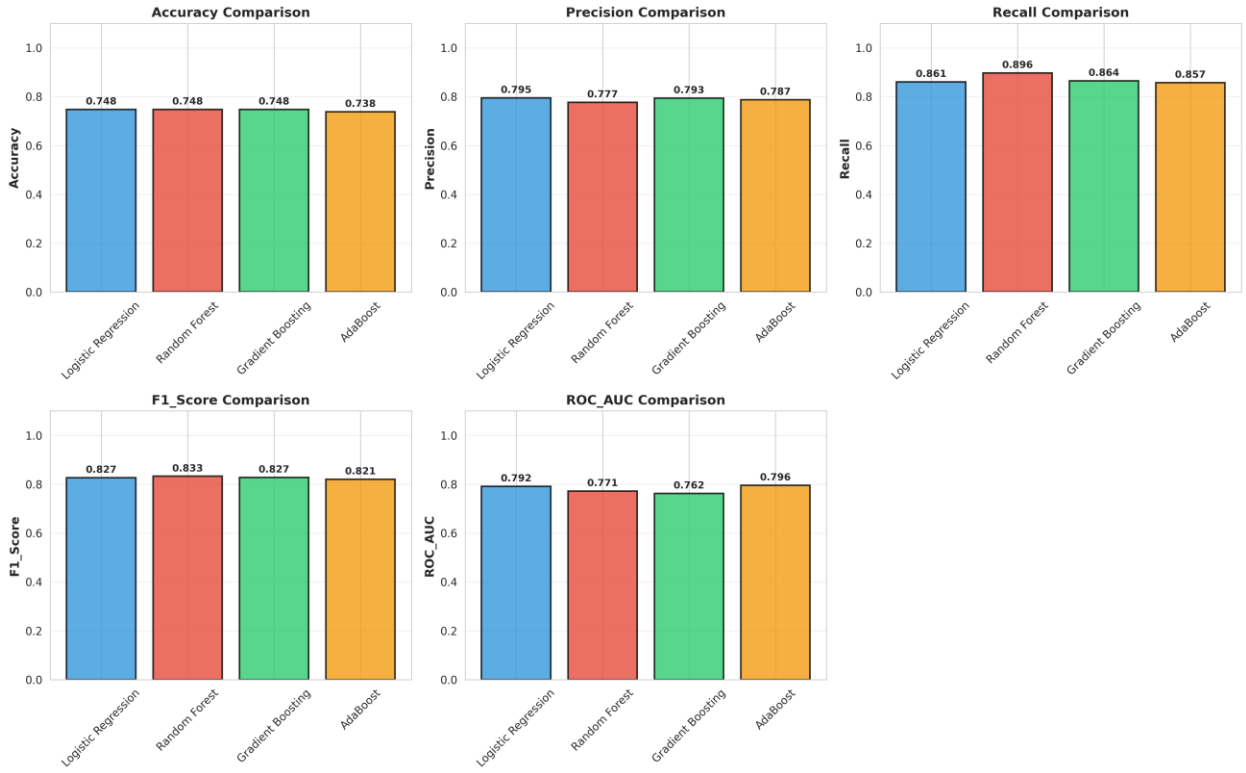


Figure 5.4: Comprehensive Model Performance Comparison

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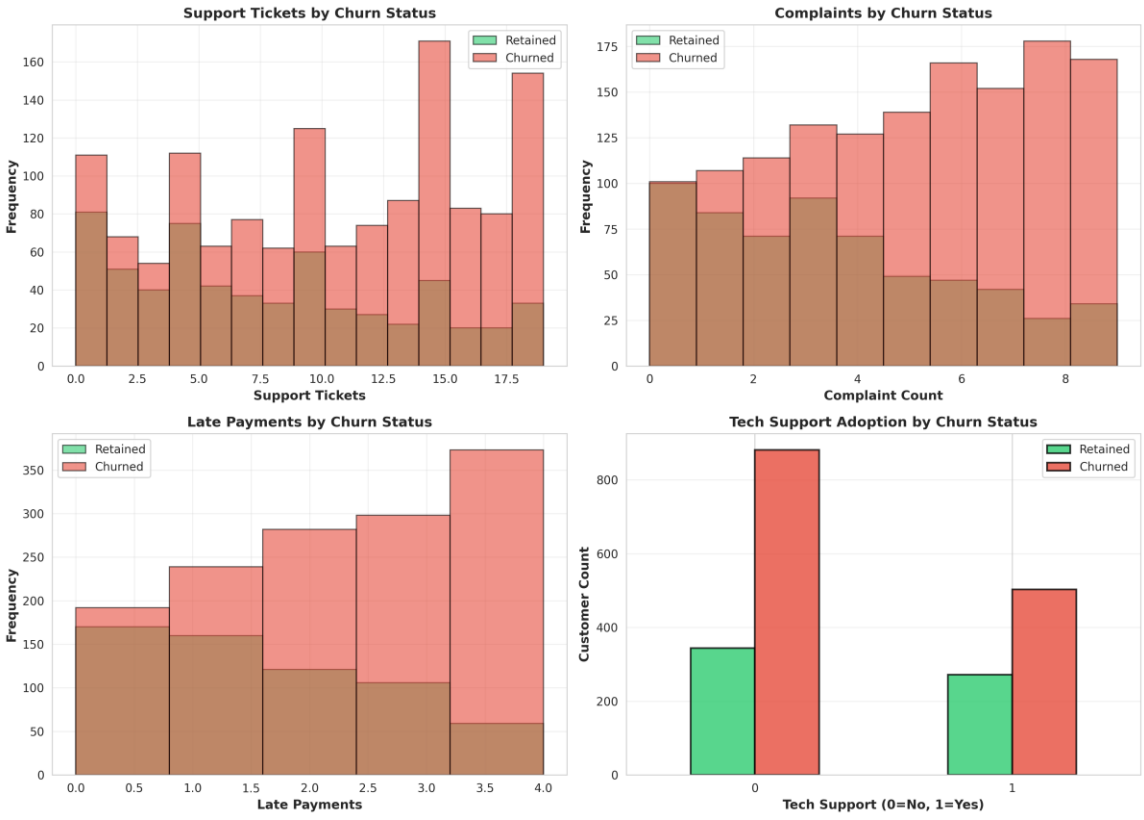


Figure 5.5: Support and Interaction Analysis Dashboard

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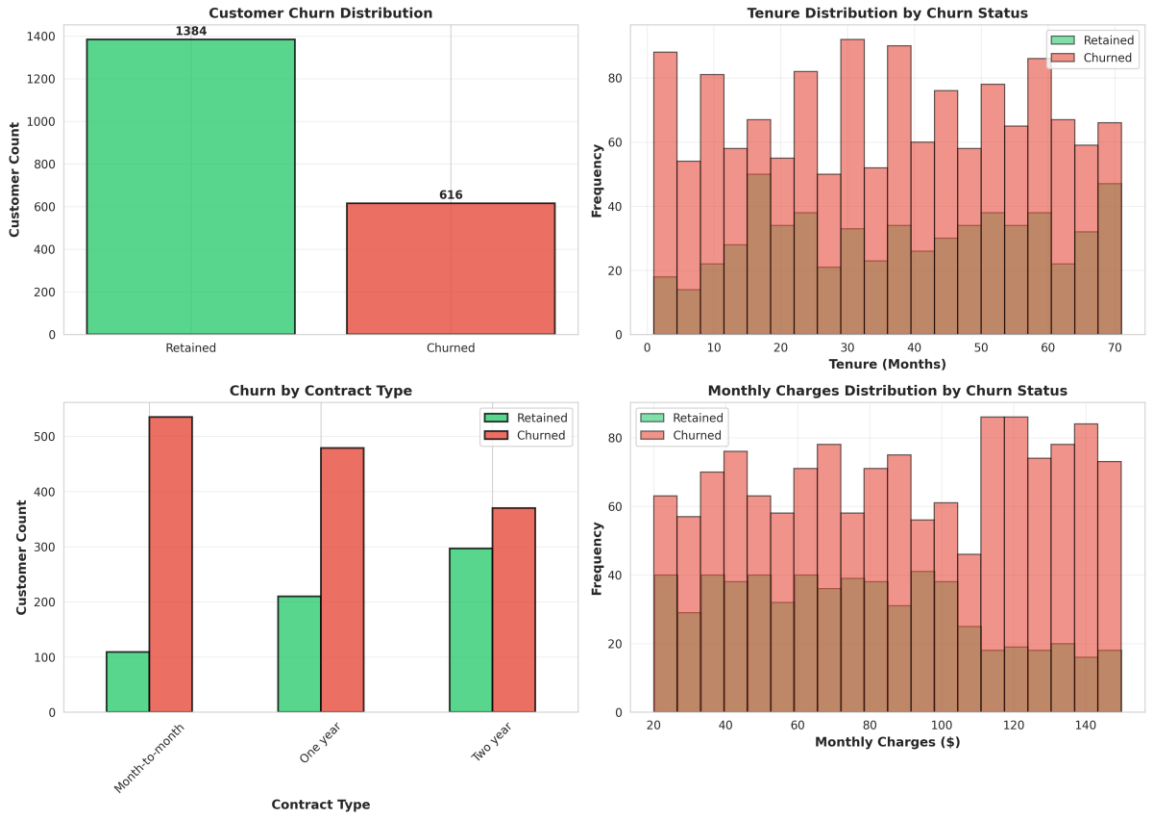


Figure 5.6: Customer Churn Distribution and Characteristics

The visualizations (Figures 5.4, 5.5, and 5.6) provide a comprehensive overview of the system's analytical capabilities. The dashboards effectively highlight trends, such as the correlation between high support tickets and increased churn rates, empowering data-driven retention management.

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CHAPTER 6

PROJECT PRESENTATION AND LEARNING EVALUATION

6.1 Technical Skill Gain

Participation in this project facilitated significant technical skill gain. The development process required mastery of Python programming, advanced data manipulation techniques using Pandas, and the practical application of machine learning classification algorithms via Scikit-learn. Furthermore, skills in data visualization and behavioral statistical analysis were substantially enhanced.

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6.2 Project Progress Review

The project progressed smoothly through defined milestones, culminating in a robust, functional predictive prototype. Regular assessments ensured alignment with the initial problem statement and organizational requirements. The successful generation of analytical datasets and comprehensive retention reports validates the project's success.

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6.3 Conclusion

Ultimately, this project delivers a modern and intelligent customer analytics solution that improves churn prediction accuracy, enhances customer retention, reduces revenue loss, and supports data-driven business decision-making. The system provides a scalable foundation for future enterprise CRM analytics initiatives.

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